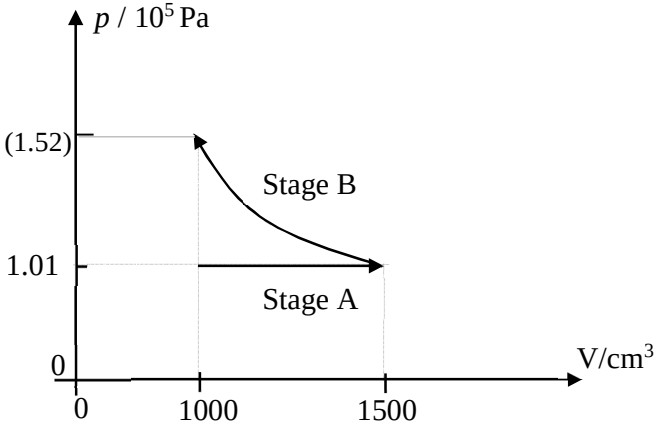


6 (a)	When a liquid is boiling, the <u>separation of its particles increases</u>. There is thus an <u>increase in microscopic potential energy</u>. There is <u>no change in temperature</u> and hence <u>no change in average microscopic kinetic energy</u>. Therefore, the <u>internal energy of the substance increases</u>. There is <u>work done by gas/against atmosphere</u> during this expansion. Since <u>W (work done on gas) is negative</u> (work done by gas is positive) and <u>ΔU is positive</u>, by reference to first law of thermodynamics, <u>Q has to be positive</u>. Therefore, thermal energy must be supplied.	[1] [1] [1] [1]
(b)(i)	The thermal <u>energy loss H</u> to the surroundings <u>is the same</u> for the duration of 5.00 minutes. $Pt = mL + H$ Hence, <u>H can be eliminated</u> (using two equations).	[1] [1]
(ii)	$(75.0)(5.00 \times 60) = (80.0 \times 10^{-3})L + H \text{ ---- (1)}$ $(20.0)(5.00 \times 60) = (23.7 \times 10^{-3})L + H \text{ ---- (2)}$ Solving, we have $L \approx 2.93 \times 10^5 \text{ J kg}^{-1}$ (not acceptable: finding the average of two different L values!)	[1] [1]

(c)(i)1	Using $pV = nRT$ Since pressure is constant, V is proportional to T for a fixed mass of gas. $V_1 / V_2 = T_1 / T_2$ $T_2 = (1500/1000) \times (273.15 + 20) = 440 \text{ K}$	[1] [1]
2.	Work done by gas = $p \times \Delta V$ $= 1.01 \times 10^5 \times (0.0015 - 0.0010)$ $= 51.0 \text{ J}$	[1]
3.	No. of moles = PV/RT $= 1.01 \times 10^5 \times 0.0010 / 8.31 \times 293.15$ $= 0.0415 \text{ moles}$	[1]
4.	Heat supplied = $mc\Delta\theta$ $= (\text{no. of moles} \times \text{molar mass}) \times 1030 \times (440 - 293.15)$ $= 0.0415 \times 0.028 \times 1030 \times (440 - 293.15)$ $= 176 \text{ J}$ Note: Deduct one mark if students did not use conversion constant of 273.15	[1] [1]
(ii)1.	Since the change is isothermal, i.e. <u>no change in temperature</u>, then there is <u>no change in internal energy</u> in stage B. This is because internal energy of an ideal gas is <u>dependent on temperature/proportional to temperature</u> only. Note: Accept answers using microscopic KE	[1] [1]
2.	 1 mark: Stage A, shape, direction and values 1 mark: Stage B, shape, direction and values Maximum 1 mark if only shape, direction of graphs	
(iii)	Since there is no change in internal energy in Stage B, the change in internal energy at the end of the 2 stage change is = change in internal energy in Stage A. By the first law of thermodynamics, the change in internal energy in Stage A is given by $\Delta U = Q + W$ $= +176 + (-51)$	[1]

	$= \underline{+125 \text{ J}}$	[1]
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7 (a)(i)	Resistance is the ratio of the potential difference across the component to	[1]
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